DT2F-TLNet Based Offline Signature Verification Using Transfer Learning and Type-2 Fuzzy Neural Networks

Abstract

Offline signature verification is an important biometric authentication technique used to distinguish genuine signatures from forged signatures. Traditional approaches often struggle to capture complex variations present in handwritten signatures. In this work, a Deep Type-2 Fuzzy Transfer Learning Network (DT2F-TLNet) is implemented for signature verification using the CEDAR signature dataset. The proposed architecture combines a pre-trained InceptionV3 network with a lightweight convolutional neural network utilizing Type-2 Fuzzy activation functions. A two-stage training strategy consisting of transfer learning and fine-tuning is employed. Experimental results demonstrate that the proposed model achieves a test accuracy of 81.44% and an AUC score of 0.8542, indicating effective discrimination between genuine and forged signatures.

1. Introduction

Handwritten signatures remain one of the most widely accepted methods of personal authentication in banking, legal documentation, and administrative processes. Manual verification is time-consuming and susceptible to human errors, motivating the development of automated signature verification systems.

Offline signature verification is particularly challenging because only the scanned image of the signature is available. Variations in writing style, pen pressure, stroke width, and skilled forgeries significantly increase the complexity of the task.

Recent advances in deep learning and transfer learning have shown promising performance in biometric verification applications. In this project, a Deep Type-2 Fuzzy Transfer Learning Network (DT2F-TLNet) is implemented to leverage the feature extraction capability of deep convolutional neural networks and the uncertainty modeling capability of Type-2 Fuzzy systems.

2. Dataset Description

CEDAR Signature Dataset

The CEDAR dataset is a publicly available benchmark dataset for offline signature verification.

Dataset characteristics:

- Total users: 55
- Genuine signatures per user: 24

- Forged signatures per user: 24
- Total images: 2640
- Genuine signatures: 1320
- Forged signatures: 1320

The dataset was divided into:

Split	Samples
Training	2112
Validation	264
Testing	264

All images were resized to 224 × 224 pixels and normalized before training.

3. Proposed Methodology

3.1 DT2F-TLNet Architecture

The proposed DT2F-TLNet architecture consists of two parallel feature extraction branches:

Model I: Type-2 Fuzzy CNN

A lightweight convolutional neural network containing multiple convolutional blocks is employed. Each convolution layer is followed by batch normalization and a custom Type-2 Fuzzy activation layer.

The fuzzy activation function helps model uncertainty and nonlinear variations present in handwritten signatures.

Model II: Transfer Learning Branch

InceptionV3 pre-trained on ImageNet is used as the transfer learning backbone. The network extracts high-level semantic features from input signature images.

Initially, all InceptionV3 layers are frozen during transfer learning.

Feature Fusion

Features extracted from both branches are concatenated and passed through fully connected layers to generate discriminative representations for signature verification.

3.2 Training Strategy

The model is trained in two phases:

Phase 1: Transfer Learning

- InceptionV3 frozen
- Only newly added layers trained
- Learning rate: 0.001

Phase 2: Fine-Tuning

- Top InceptionV3 layers unfrozen
- Learning rate reduced to 1e-5
- Fine-tuning performed to improve feature adaptation

4. Experimental Setup

Experiments were performed using:

- Google Kaggle Environment
- NVIDIA Tesla GPU
- TensorFlow / Keras
- Python

Training parameters:

Parameter	**Value**
Image Size	224 × 224
Batch Size	32
Optimizer	RMSprop
Loss Function	Binary Crossentropy
Epochs	15 (Phase 1) + Fine-Tuning
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score, AUC

Parameter **Value**

```
DT2F-TLNet
============================================================
Total Params: 23,411,050
Trainable Params: 1,602,778
Non-Trainable Params: 21,808,274
Model: "DT2F_TLNet"
```

Layer (type)	Output Shape	Param #	Connected to
input_layer_14 (InputLayer)	(None, 224, 224, 3)	0	-
conv2d_774 (Conv2D)	(None, 224, 224, 16)	432	input_layer_14[0…
batch_normalizatio… (BatchNormalizatio…	(None, 224, 224, 16)	64	conv2d_774[0][0]
type2_fuzzy_26 (Type2Fuzzy)	(None, 224, 224, 16)	48	batch_normalizat…
max_pooling2d_48 (MaxPooling2D)	(None, 112, 112, 16)	0	type2_fuzzy_26[0…
conv2d_775 (Conv2D)	(None, 112, 112, 32)	4,608	max_pooling2d_48…
batch_normalizatio… (BatchNormalizatio…	(None, 112, 112, 32)	128	conv2d_775[0][0]
type2_fuzzy_27 (Type2Fuzzy)	(None, 112, 112, 32)	96	batch_normalizat…
max_pooling2d_49 (MaxPooling2D)	(None, 56, 56, 32)	0	type2_fuzzy_27[0…
conv2d_776 (Conv2D)	(None, 56, 56, 64)	18,432	max_pooling2d_49…
batch_normalizatio… (BatchNormalizatio…	(None, 56, 56, 64)	256	conv2d_776[0][0]
type2_fuzzy_28 (Type2Fuzzy)	(None, 56, 56, 64)	192	batch_normalizat…
max_pooling2d_50 (MaxPooling2D)	(None, 28, 28, 64)	0	type2_fuzzy_28[0…
conv2d_777 (Conv2D)	(None, 28, 28, 128)	73,728	max_pooling2d_50…
batch_normalizatio… (BatchNormalizatio…	(None, 28, 28, 128)	512	conv2d_777[0][0]
type2_fuzzy_29 (Type2Fuzzy)	(None, 28, 28, 128)	384	batch_normalizat…
max_pooling2d_51 (MaxPooling2D)	(None, 14, 14, 128)	0	type2_fuzzy_29[0…
conv2d_778 (Conv2D)	(None, 14, 14, 256)	294,912	max_pooling2d_51…
inception_v3 (Functional)	(None, 5, 5, 2048)	21,802,784	input_layer_14[0…
batch_normalizatio… (BatchNormalizatio…	(None, 14, 14, 256)	1,024	conv2d_778[0][0]
global_average_poo… (GlobalAveragePool…	(None, 2048)	0	inception_v3[0][…
type2_fuzzy_30 (Type2Fuzzy)	(None, 14, 14, 256)	768	batch_normalizat…

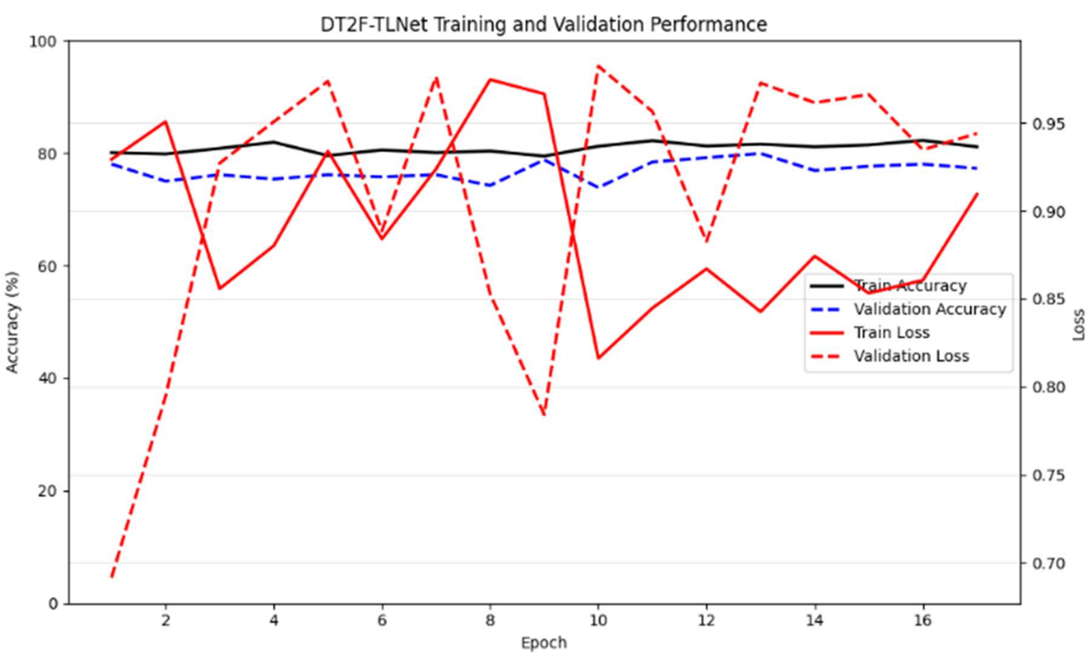
Parameter **Value**

batch_normalizatio… (BatchNormalizatio…	(None, 2048)	8,192	global_average_p…
global_average_poo… (GlobalAveragePool…	(None, 256)	0	type2_fuzzy_30[0…
dense_12 (Dense)	(None, 512)	1,049,088	batch_normalizat…
concatenate_20 (Concatenate)	(None, 768)	0	global_average_p… dense_12[0][0]
dense_13 (Dense)	(None, 200)	153,600	concatenate_20[0…
batch_normalizatio… (BatchNormalizatio…	(None, 200)	800	dense_13[0][0]
type2_fuzzy_31 (Type2Fuzzy)	(None, 200)	600	batch_normalizat…
dropout_4 (Dropout)	(None, 200)	0	type2_fuzzy_31[0…
dense_14 (Dense)	(None, 2)	402	dropout_4[0][0]

Total params: 23,411,050 (89.31 MB)
Trainable params: 1,602,778 (6.11 MB)
Non-trainable params: 21,808,272 (83.19 MB)

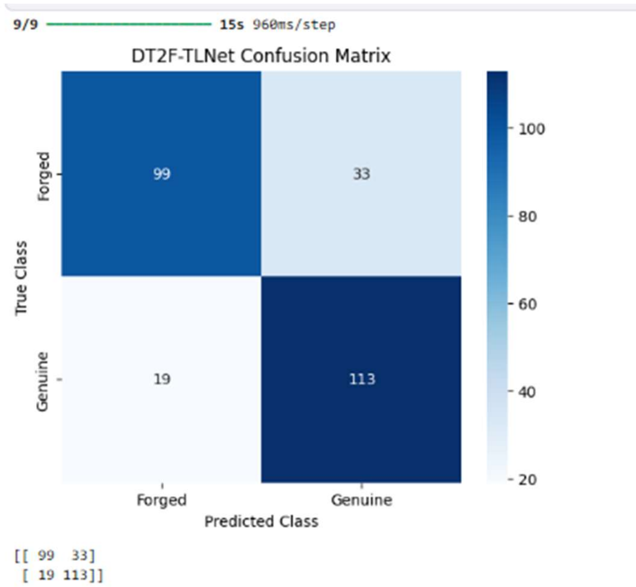
5. Results and Discussion

5.1 Training Performance



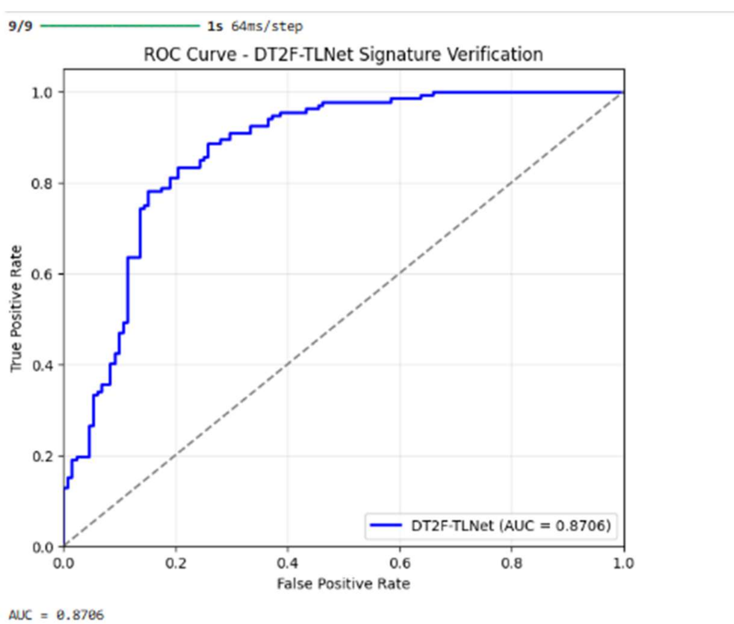
The training and validation curves demonstrate stable convergence of the DT2F-TLNet model. Fine-tuning further improves feature learning while preventing significant overfitting.

5.2 Confusion Matrix



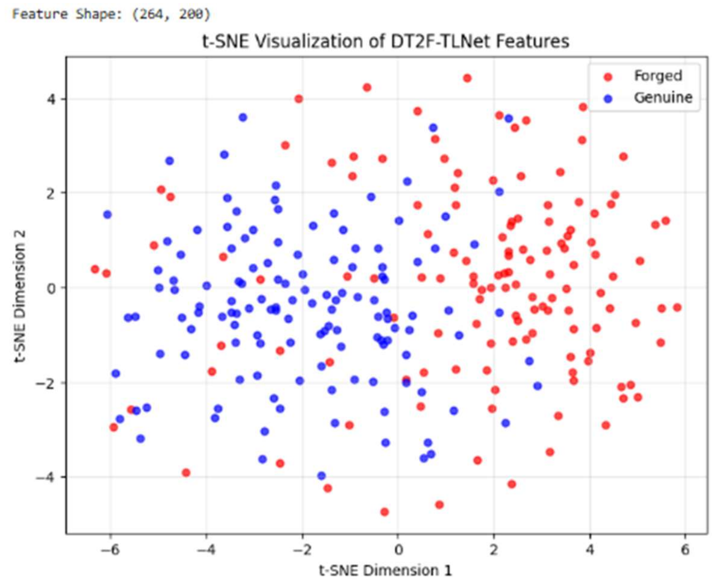
The confusion matrix shows that the majority of genuine and forged signatures are correctly classified. Misclassification occurs mainly in difficult forged samples that closely resemble genuine signatures.

5.3 ROC Analysis



The ROC curve demonstrates strong separation between genuine and forged signatures. The obtained AUC score confirms the effectiveness of the learned feature representations.

5.4 t-SNE Visualization



The t-SNE visualization illustrates the feature space learned by DT2F-TLNet. Genuine and forged signatures form distinguishable clusters, indicating that the proposed model successfully learns discriminative representations.

6. Performance Evaluation

Phase 1 Results

Metric	**Value**
Validation Accuracy	80.30%
Validation AUC	0.8549

Phase 2 Results

Metric	**Value**
Validation Accuracy	79.92%
Validation AUC	0.8763

Final Test Results

Metric	**Value**
Accuracy	81.44%
Precision	77.40%
Recall	85.61%

Metric	Value
F1-Score	81.29%
AUC	0.8542

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9/9 ━━━━━━━━━━━━━━━━━━━━ 1s 65ms/step
============================================================
TEST METRICS
============================================================
Accuracy : 0.803030303030303
Precision: 0.773972602739726
Recall   : 0.8560606060606061
F1 Score : 0.8129496402877698

Classification Report
              precision    recall  f1-score   support

      Forged       0.84      0.75      0.79       132
     Genuine       0.77      0.86      0.81       132

    accuracy                           0.80       264
   macro avg       0.81      0.80      0.80       264
weighted avg       0.81      0.80      0.80       264
```

The results indicate that fine-tuning improves the generalization capability of the model and enhances its ability to distinguish between genuine and forged signatures.

7. Conclusion

In this work, a DT2F-TLNet architecture combining Type-2 Fuzzy convolutional layers and transfer learning using InceptionV3 was successfully implemented for offline signature verification. The model was trained using a two-stage strategy consisting of transfer learning and fine-tuning.

Experimental evaluation on the CEDAR signature dataset demonstrated that the proposed approach effectively learns discriminative features for signature verification. The final model achieved a test accuracy of 81.44% and an AUC score of 0.8542.

Future work may include exploring larger signature datasets, advanced data augmentation techniques, attention mechanisms, and transformer-based architectures to further improve verification performance.